A NOETHERIAN INTEGRAL SEPARATED SCHEME OF DIMENSION 1 HAS AN AMPLE INVERTIBLE SHEAF

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Both invoked lemmas are delivered companions of this lane; the proof's own content is the finite-complement observation and the assembly.

Allowed standard prerequisites (editorial).

- Producing globally generated pairs with prescribed nonvanishing loci. delivered companion lemma-find-globally-generated. Gives (L_i, s_i) with U_i = X_{s_i}.
- Enough globally generated invertible sheaves give an ample one. delivered companion lemma-enough-globally-generated-ample. Final conclusion.

Source reference. Original lemma 09NB: sources/stacks-chow/tags.txt, source lines 12209–12209. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09NC: sources/stacks-chow/tags.txt, source lines 12210–12210. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a Noetherian integral separated scheme of dimension 1. Then X has an ample invertible sheaf.*

Proof. Choose an affine open covering $X = U_1 \cup \ldots \cup U_n$. Since X is Noetherian, each of the sets $X \setminus U_i$ is finite. Thus by Lemma 09NB we can find a pair $(\mathcal{L}_i, s_i)$ consisting of a globally generated invertible sheaf $\mathcal{L}_i$ and a global section s_i such that $U_i = X_{s_i}$. We conclude that X has an ample invertible sheaf by Lemma 09NC. □

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